3.31 Discrete Valuation Rings

A Discrete Valuation Ring (DVR) corresponds to a regular local ring of dimension 1 and therefore regular (or in the case of perfect base field, non-singular) points on a curve. There are a number of equivalent conditions which are summarised here.

Lemma 3.31.1
Let $(A, \mathfrak{m})$ be a local domain such that $\mathfrak{m}$ is finitely generated and $\dim A = 1$. Then for every non-zero ideal $\mathfrak{a}$ there exists some n such that $\mathfrak{m}^n \subseteq \mathfrak{a}$.

Proof. As $\dim A = 1$ every minimal prime over $\mathfrak{a}$ is equal to $\mathfrak{m}$, whence $\sqrt{\mathfrak{a}} = \mathfrak{m}$ (i.e. $\mathfrak{a}$ is $\mathfrak{m}$-primary). As $\mathfrak{m}$ is finitely generated there is some n such that $\mathfrak{m}^n \subseteq \mathfrak{a}$ (3.3.47). ∎

Proposition 3.31.2 (DVR Criteria)
Let A be an integral domain which is not a field. Then the following are equivalent.

a) A is a valuation ring whose valuation group is order-isomorphic to $\mathbb{Z}$
b) A is a local principal ideal domain
c) A is a Noetherian local ring with maximal ideal $\mathfrak{m} = (\pi)$ and $\dim A = 1$
d) A is a local ring such that every ideal is of the form (π^k) for some $k \geq 0$ (and these are distinct)
e) There exists a non-unit $\pi \in A$ such that every element $a \in A$ may be represented as $u\pi^k$ for some $u \in A^\star$ and non-negative integer k.
f) A is a Noetherian valuation ring
g) A is a Noetherian integrally closed local domain with $\dim A = 1$
h) A is a Noetherian regular local ring with $\dim A = 1$

*In this case we say A is a **discrete valuation ring** (or **DVR**) and let $K = \mathrm{Frac}(A)$. For every generator π of $\mathfrak{m}$, the ideals (π^n) are distinct and*

- *for every $x \in K^\star$ there is a unique integer k and unit u such that $x = u\pi^k$, and*
- *there is a well-defined valuation $v : K^\star \to \mathbb{Z}$ determined uniquely by the relation $x = u\pi^{v(x)}$*

Proof. a) $\implies$ b) Consider a valuation $v : K \to \mathbb{Z}$ with $v^{-1}(\mathbb{Z}_{\geq 0}) = A$. For any ideal $\mathfrak{a}$ choose $f \in \mathfrak{a}$ such that $v(f)$ is minimal. For $g \in \mathfrak{a}$ we have $v(g) \geq v(f) \implies v(gf^{-1}) \geq 0 \implies gf^{-1} \in A \implies g \in (f)$. Therefore $\mathfrak{a} = (f)$ is principal. Further by (3.25.2) A is a local ring.

b) $\implies$ c). By (3.22.4) $\dim A = 1$ and by assumption $\mathfrak{m} = (\pi)$. Evidently A is Noetherian.

c) $\implies$ d). By (3.29.1) the sequence (π^k) is strictly decreasing. Assume wlog that $\mathfrak{a}$ is proper and non-zero, then as $\dim A = 1$ every prime ideal is maximal and we find $\sqrt{\mathfrak{a}} = (\pi)$ by (3.3.45).

Suppose that $\mathfrak{a} \subseteq (\pi^k)$ for all k. Then $\pi \in \sqrt{\mathfrak{a}} \implies (\pi^k) = \mathfrak{a}$ for all sufficiently large k which contradicts the fact the sequence is strictly decreasing. Therefore we may assume $\mathfrak{a} \subseteq (\pi^k)$ for some k and there exists $a \in \mathfrak{a} \setminus (\pi^{k+1})$. Therefore $a = u\pi^k$ and by construction $u \notin (\pi) \implies u \in A^\star \implies \pi^k \in \mathfrak{a}$. Therefore $\mathfrak{a} = (\pi^k)$ as required.

d) $\implies$ e). By assumption $(a) = (\pi^k)$ for some integer k, whence by (3.17.3) $a = u\pi^k$ for some $u \in A^\star$. This representation is unique because the ideals (π^k) are all distinct and A is cancellable.

e) $\implies$ a). We may show easily that the integer k is uniquely defined, denote this by $v(a)$ where $a = u\pi^{v(a)}$ for $u \in A^\star$. We need to show that v is a valuation. By uniqueness we deduce $v(ab) = v(a) + v(b)$. Suppose that $a, b, a + b \in A \setminus \{0\}$. Then $(\pi^{v(a+b)}) = (a + b) \subseteq (\pi^{\min(v(a),v(b))})$. As the sequence (π^k) is strictly decreasing we find $v(a + b) \geq \min(v(a), v(b))$ as required. We may extend to $K^\star$ by $v(ab^{-1}) := v(a) - v(b)$. Then $v(ab^{-1}) \geq 0 \iff v(a) \geq v(b) \iff (a) \subseteq (b) \iff b \mid a \iff ab^{-1} \in A$. Further $v(a) = 0 \iff a \notin (\pi) \iff a \in A^\star$. Therefore A is the valuation ring associated to v.

d) $\implies$ f) $\implies$ b). This follows from (3.25.20).

f) $\implies$ g). A valuation ring is integrally closed (3.25.2) and we've already shown A is local with dimension 1.

g) $\implies$ c). Let $0 \neq a \in \mathfrak{m}$. As $\dim A = 1$ from (3.31.1) we know $\mathfrak{m}^n \subseteq (a)$ for some $n \geq 1$. Choose n minimal and $b \in \mathfrak{m}^{n-1} \setminus (a)$.

Let $x = a^{-1}b \in K = \mathrm{Frac}(A)$. Evidently $x \notin A$, for otherwise $b \in (a)$; further $x\mathfrak{m} = a^{-1}b\mathfrak{m} \subseteq a^{-1}\mathfrak{m}^n \subseteq a^{-1}(a) = A$ is an ideal of A.

We claim that $x\mathfrak{m} = A$. Suppose not then $x\mathfrak{m} \subseteq \mathfrak{m}$ by (3.3.36). Then $\mathfrak{m}$ is a faithful finite $A[x]$-module and x is integral over A (3.24.3). By assumption A is integrally closed so $x \in A$ which is a contradiction. Therefore $x\mathfrak{m} = A \implies \mathfrak{m} = x^{-1}A$. This shows that $x^{-1} = x^{-1} \cdot 1 \in A$ and that $\mathfrak{m}$ is a principal ideal.

c) $\implies$ h). By assumption A is a Noetherian local ring with $\dim A = 1$. So it remains to show that $\dim \mathfrak{m}/\mathfrak{m}^2 = 1$. As $\mathfrak{m}$ is principal then $\dim \mathfrak{m}/\mathfrak{m}^2 \leq 1$. If it is zero then $\mathfrak{m} = \mathfrak{m}^2$ and $\dim A = 0$ by (3.29.1), a contradiction.

h) $\implies$ c). By assumption $\dim \mathfrak{m}/\mathfrak{m}^2 = 1$ and so by (3.21.7) $\mathfrak{m} = (\pi)$ for some $\pi \in \mathfrak{m} \setminus \mathfrak{m}^2$.

The final statements have already been proven. ∎

Lemma 3.31.3 (Valuation and Powers of the Maximal Ideal)
Let $(A, \mathfrak{m})$ be a DVR with valuation v, and $k \geq 0$ an integer. For $0 \neq a \in A$:

a) $v(a) \geq k \iff a \in \mathfrak{m}^k$;
b) $v(a) = k \iff a \in \mathfrak{m}^k \setminus \mathfrak{m}^{k+1}$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Fix a uniformiser π (so $\mathfrak{m} = (\pi)$, hence $\mathfrak{m}^k = (\pi^k)$ (3.31.2)), and write $a = u\pi^{v(a)}$ for a unit $u \in A^\star$ (3.31.2).

a). $a \in \mathfrak{m}^k = (\pi^k) \iff \pi^k \mid a = u\pi^{v(a)} \iff \pi^k \mid \pi^{v(a)}$ (as u is a unit) $\iff v(a) \geq k$ (as the ideals (π^n) are strictly decreasing (3.31.2)).

b). By a), $a \in \mathfrak{m}^k \setminus \mathfrak{m}^{k+1} \iff v(a) \geq k$ and not $(v(a) \geq k+1) \iff v(a) = k$. ∎

Corollary 3.31.4
Let A be a principal ideal domain which is not a field. Then every localisation of A at a non-zero prime ideal is a DVR.

In particular $\mathbb{Z}_{(p)}$ and $k[X]_{(f)}$ are DVRs where p is prime and $f(X)$ is an irreducible polynomial.

Lemma 3.31.5 (A DVR is Maximal Among Valuation Subrings of its Fraction Field)
Let A be a DVR with fraction field K and B a valuation ring of K with $A \subseteq B$. Then $A = B$ or $B = K$.

Proof. Let π be a uniformiser of A, so every $a \in A$ is $u\pi^k$ for a unit $u \in A^\star$ and integer $k \geq 0$ (3.31.2). Suppose $B \neq K$. As B is a valuation ring of K, either $\pi \in B$ (already true, as $\pi \in A \subseteq B$) or $\pi^{-1} \in B$; we rule out the latter. If $\pi^{-1} \in B$, then for any $x \in K^\star$, writing $x = u\pi^n$ ($u \in A^\star \subset B$, $n \in \mathbb{Z}$, using A a DVR), $x = u\pi^n \in B$ whether $n \geq 0$ (as $\pi^n \in A \subset B$) or $n < 0$ (as $\pi^n = (\pi^{-1})^{-n} \in B$); so $B = K$, contradicting assumption. Hence $\pi^{-1} \notin B$, that is $\pi \in \mathfrak{m}_B$. Then $\mathfrak{m}_A = \pi A \subset \pi B \subset \mathfrak{m}_B$ (as $\pi \in \mathfrak{m}_B$ and B absorbs), so B dominates A; as A is a valuation ring, it is maximal under domination (3.25.13), giving $A = B$. ∎

Lemma 3.31.6 (Directed Union of Valuation Rings with a Common Uniformiser)
Let $(K_i)_{i \in I}$ be a directed system of fields with union $K := \bigcup_i K_i$, and for each i let $R_i \subset K_i$ be a valuation ring with $R_i \subseteq R_l$ whenever $K_i \subseteq K_l$. Suppose in addition every $\mathfrak{m}_{R_i}$ is principal, generated by a single common element $0 \neq \pi \in \bigcap_i R_i$. Then $R := \bigcup_i R_i$ is a valuation ring of K, with $\mathfrak{m}_R = \pi R \neq 0$; in particular $R \subsetneq K$.

Proof. For $K_i \subseteq K_l$ (so $R_i \subseteq R_l$), $\mathfrak{m}_{R_i} = \pi R_i \subseteq \pi R_l = \mathfrak{m}_{R_l}$, giving R_l dominates R_i (3.25.4) so (3.25.7) applies, and $\mathfrak{m}_R = \bigcup_i \mathfrak{m}_{R_i} = \bigcup_i \pi R_i = \pi R$.

As $\pi \in \mathfrak{m}_{R_i}$ is a non-unit of every R_i, $\pi^{-1} \notin R_i$ for every i (well-defined, as $0 \neq \pi \in K_i \subset K$), hence $\pi^{-1} \notin R = \bigcup_i R_i$; as $\pi^{-1} \in K$, $R \subsetneq K$. ∎

Proposition 3.31.7
Let A be a discrete valuation ring with unique maximal ideal $\mathfrak{m}$. Then there exists a canonical group isomorphism

$$(\mathfrak{m})^- : \mathbb{Z} \xrightarrow{\sim} \mathcal{I}_A$$

which is the unique group homomorphism mapping 1 to $\mathfrak{m}$ (and 0 to A. In particular every fractional ideal is invertible.

Write the inverse group isomorphism by $v_\mathfrak{m}$, which satisfies

a) $\mathfrak{m}^{v_\mathfrak{m}(\mathfrak{a})} = \mathfrak{a}$
b) $v_\mathfrak{m}(\mathfrak{m}^k) = k$

A fractional ideal I is contained in A if and only if $v_\mathfrak{m}(I) \geq 0$.

Proof. By (3.31.2) we may suppose $\mathfrak{m} = (\pi)$. Evidently (π^k) is a fractional ideal for every integer k and $(\pi^k)(\pi^l) = (\pi^{k+l})$, which gives the required homomorphism $k \to (\pi^k)$.

Suppose the kernel is non-trivial then there exists k which maps to A, and we may assume that k is positive (by considering the inverse). This means precisely that $\mathfrak{m}^k = A$, which may only happen when $k = 0$ because $\mathfrak{m}$ is proper.

Let I be an arbitrary fractional ideal then $aI \subseteq A$ for some $a \in A$. By (3.31.2) $a = u\pi^l$ and $aI = (\pi^k)$ for some positive integers l and k. This means $(\pi^l)I = (\pi^k)$ and therefore $I = (\pi^k)(\pi^{-l}) = (\pi^{k-l})$. This shows the homomorphism is surjective and therefore an isomorphism.

Evidently $(\pi^k) \subset A \iff k \geq 0$ since $\pi \in A$ and $\pi^{-1} \notin A \implies \pi^{-k} \notin A$. ∎

Proposition 3.31.8
Let A be a discrete valuation ring and I, J fractional ideals. Then the following are equivalent

a) $I \subset J$
b) J divides I (that is there exists an ideal $\mathfrak{a}$ such that $\mathfrak{a}J = I$)
c) $v_{\mathfrak{m}}(I) \geq v_{\mathfrak{m}}(J)$

Proof. Using the results of (3.31.7) we may show (eliding the subscript $\mathfrak{m}$)

a) $\iff$ c) $I \subset J \iff J^{-1}I \subset A \iff v(J^{-1}I) \geq 0 \iff v(I) \geq v_{\mathfrak{m}}(J)$

b) $\implies$ c) $v(I) = v(\mathfrak{a}) + v(J) \geq v(J)$

c) $\implies$ b) Define $\mathfrak{a} = \mathfrak{m}^{v(I)-v(J)}$ then evidentally $v(\mathfrak{a}J) = v(I)$ whence $I = J$. ∎

Lemma 3.31.9 (Intersection of the Powers of the Maximal Ideal)
Let A be a DVR with maximal ideal $\mathfrak{m}$. Then $\bigcap_{n\geq 0} \mathfrak{m}^n = 0$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Let $0 \neq a \in A$, and write $a = u\pi^{v(a)}$ for a unit u and uniformiser π (3.31.2). If $a \in \mathfrak{m}^{v(a)+1} = (\pi^{v(a)+1})$ then $a = u'\pi^{v(a)+1}$ for some $u' \in A$, so $u\pi^{v(a)} = u'\pi^{v(a)+1}$, giving $u = u'\pi$ by cancellation – contradicting u a unit. So $a \notin \mathfrak{m}^{v(a)+1}$, in particular $a \notin \bigcap_n \mathfrak{m}^n$. ∎

Lemma 3.31.10 (Ramification and Injectivity/Surjectivity of Local Homomorphisms)
*Let $(A, \mathfrak{m})$, $(B, \mathfrak{n})$ be DVRs with valuations v_A, v_B (3.31.2), and $\phi : A \to B$ a local ring homomorphism (that is $\phi(\mathfrak{m}) \subseteq \mathfrak{n}$). Fix a uniformiser π of A and set $e := v_B(\phi(\pi)) \in \mathbb{N} \cup \{\infty\}$ (the **ramification index** of ϕ; independent of the choice of π, with $e = \infty$ iff $\phi(\pi) = 0$). Then:*

a) ϕ is injective if and only if $e < \infty$;
b) if ϕ is surjective, then $e = 1$; in particular, by a), ϕ is then also injective – so ϕ is surjective if and only if it is bijective.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Independence of π. If $\pi' = u\pi$ for a unit $u \in A^\star$, then $\phi(\pi') = \phi(u)\phi(\pi)$ with $\phi(u) \in B^\star$ (units map to units under any ring homomorphism, and $\phi(u) \notin \mathfrak{n}$, as otherwise $1 = \phi(u^{-1})\phi(u) \in \mathfrak{n}$, absurd), so $v_B(\phi(\pi')) = v_B(\phi(\pi))$.

a). $\ker\phi$ is a prime ideal of A (the kernel of a ring homomorphism into the domain B), so $\ker\phi \in \{0, \mathfrak{m}\}$ (3.31.2). If $e = \infty$: $\phi(\pi) = 0$, so $\pi \in \ker\phi$, and, as $\mathfrak{m} = (\pi)$, $\mathfrak{m} \subseteq \ker\phi$; as $\ker\phi \neq A$ ($\phi(1) = 1 \neq 0$), $\ker\phi = \mathfrak{m} \neq 0$: ϕ is not injective. If $e < \infty$: $\phi(\pi) \neq 0$, so $\pi \notin \ker\phi$, giving $\ker\phi \neq \mathfrak{m}$, so $\ker\phi = 0$: ϕ is injective.

b). Suppose ϕ is surjective. Let ϖ be a uniformiser of B; by surjectivity, $\varpi = \phi(a)$ for some $a \in A$, necessarily $a \neq 0$ (as $\varpi \neq 0$). Write $a = u\pi^k$ for a unit $u \in A^\star$ and $k := v_A(a) \geq 0$ (3.31.2). If $e = \infty$: $\phi(a) = \phi(u)\phi(\pi)^k$ is $\phi(u)$ (a unit, as units map to units) if $k = 0$, or 0 if $k \geq 1$; neither equals ϖ ($v_B(\varpi) = 1 \neq 0$, $\varpi \neq 0$), a contradiction. So $e < \infty$, and

$$1 = v_B(\varpi) = v_B(\phi(u)) + k \cdot v_B(\phi(\pi)) = 0 + ke = ke,$$

forcing $e = 1$. ∎